



ZUS801LP Environmental Sensor Instructions Manual

Complete setup guide: hardware connection, serial CLI, network
& MQTT configuration, and Grafana dashboard access

Model	Sensor	Firmware	Revision
ZUS801LP	8-in-1 · BME688	v1.1.3+	A01

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ZUS801LP Technical Specifications

ZUS801LP — 8-in-1 SPE PoDL/SPoE Industrial Environmental Sensor

Measurements	Temperature · Pressure · Humidity · Air Quality (VOC/CO ₂ eq) · Ambient Light · Ambient Sound · Proximity · 3-Axis Acceleration
Connectivity	1 × 10BASE-T1L · IEC 63171-1 connector · IEEE 802.3cg
Distance	Up to 1,700 m at 10 Mb/s
Power input	PoDL/SPoE Class 13 (50–58 VDC) or Class 10/11 (20–30 VDC)
Protocol	MQTT (Publisher) · TLS 1.2 and 1.3
Security	Secure Element (SE) · Crypto accelerator · Secure key storage · RNG
Operating temp	–40 to +70 °C
Humidity (operating)	5 to 95 %RH non-condensing
Dimensions	51 × 89 × 25 mm (2 × 3.5 × 1 in)
Weight	0.1 kg (0.22 lb)
Housing	Rugged ABS plastic
Mounting	¼" standard camera screw attachment
LEDs	Port LED (Link/Activity) · Multicolor status LED
Standard	IEEE 802.3cg (10BASE-T1L)

Required Hardware

Item	Specification	Notes
Zemfyre ZUS801LP Sensor	8-in-1 industrial environmental	SPE/PoDL connectivity
Raspberry Pi (server host)	Pi 3 or newer	Runs Docker stack
SPE cable	Per IEC 63171-1 — up to 1,700 m	Single Pair Ethernet to sensor
PoDL/SPoE power source	Class 13 (50–58 VDC) or Class 10/11 (20–30 VDC)	Powers sensor over cable
USB-to-Serial adapter	CH340 / CP210x / FTDI	For direct CLI access
PC / Laptop	Windows / macOS / Linux	For initial configuration

Physical Connections

- 1 Power:** Connect the ZUS801LP to a suitable power supply and verify the power LED lights up.
- 2 SPE Ethernet:** Connect the sensor to the network using a Single Pair Ethernet (SPE) cable.
- 3 Serial (optional):** Connect a USB-to-Serial adapter between the sensor UART header and your PC. Install the driver for your adapter (CH340, CP210x, or FTDI).

Virgin Board State

A freshly-programmed sensor has an unprogrammed EEPROM. The status LED toggles **YELLOW**

, indicating no MQTT broker is configured. Sensors do not initialize until MQTT binding is set up.

2 Initial Connection

2.1 Serial Connection via USB-to-Serial Adapter

How the connection works

The ZUS801LP exposes a

UART header

(TX/RX/GND pins) — not a native USB port. A USB-to-Serial adapter (CH340, CP210x, or FTDI chip) bridges that UART to your PC's USB port and appears in Windows Device Manager as a

Virtual COM Port

(e.g. **COM4**). This is the adapter you configure in PuTTY, not a direct USB cable.

PuTTY Settings

Setting	Value	Description
Connection type	Serial	Direct serial communication
Serial line	COM4	Device Manager → Ports → USB Serial Device (COMx)
Speed (baud)	115200	Standard sensor baud rate
Data bits	8	Default
Stop bits	1	Default
Flow control	None	Default

Finding Your COM Port

- 1
- Plug the USB-to-Serial adapter into your PC.
- 2
- Open **Device Manager** (`Win + X` → Device Manager) → expand **Ports (COM & LPT)** .
- 3
- Note the port shown — e.g. *USB Serial Device (COM4)* . Use that number in PuTTY.
- 4
- Launch PuTTY, select **Serial** , enter the COM port number and baud **115200** .
- 5
- Optionally save the session as *"zemfyr-sensor"* . Click **Open** ; `Enter` if no prompt appears.

Driver Required

If no COM port appears in Device Manager, install the adapter driver first: CH340 → *ch341ser.exe* · CP210x → *Silicon Labs CP210x VCP* · FTDI → *CDM FTDI drivers*

2.2 Network Connection

After network settings have been configured and a system reset performed, the CLI can also be accessed remotely (Telnet/SSH) at the sensor's assigned IP address.

Prerequisite

Network-based CLI access requires IP configuration via the serial connection first — see Section 3.

3 Network Configuration

IP addressing convention used in this manual

Raspberry Pi (MQTT broker, eth0): `192.168.2.30` · Sensors: `192.168.2.100` , `192.168.2.101` , `192.168.2.102` ... · Gateway: `192.168.2.1` . Adapt to your network as needed — the sensor and broker must be on the same subnet.

Configure Raspberry Pi Host Interface

```
sudo nmcli connection modify eth0 ipv4.method manual ipv4.addresses "192.168.2.30/24"
sudo nmcli connection up eth0
ip -4 addr show eth0
```

View & Set Sensor Network Binding

```
# Show current network settings
eth -ifconfig

# Set device IP address (first sensor = .100, second = .101, etc.)
eth -ifconfig ip 192.168.2.100

# Set gateway
eth -ifconfig gw 192.168.2.1

# Set subnet mask
eth -ifconfig sn 255.255.255.0

# Set DNS servers (optional)
eth -ifconfig dns0 8.8.8.8
eth -ifconfig dns1 8.8.4.4

# Verify all settings
eth -ifconfig
```

System Reset Required

Network changes are written to EEPROM but do not take effect until: `core -reset`

Apply Changes & Verify

```
core -reset

# Test connectivity after reboot
eth -ping 192.168.2.100
```

After reset, the sensor comes back online with the new network settings. The LED changes to  **green toggling** once MQTT is also configured.

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MQTT Broker Setup

Verify Mosquitto on the Host

```
docker ps | grep mosquitto
ss -ltnp | grep 1883
mosquitto_sub -h 127.0.0.1 -p 1883 -t '#' -v
```

Configure MQTT on the Sensor

```
# View current MQTT binding
eth -mqtt

# Set broker IP address
eth -mqtt ip 192.168.2.30

# Set broker port
eth -mqtt port 1883

# Alternative: set by hostname/URL
eth -mqtt url "mqtt-broker.local"

# Verify configuration
eth -mqtt

# Apply changes
core -reset

# Monitor publish statistics
eth -mqtt stats
```

LED Status Reference

LED State	Meaning
● Yellow — toggling	MQTT broker not configured or unreachable
● Green — toggling	Broker configured; connection attempt in progress
● Green — solid	Connected; subscription active; data flowing

Success Indicator

Solid green LED means the sensor is fully operational. Data should now appear in InfluxDB and the Grafana dashboard.

Security

The ZUS801LP supports

TLS 1.2 and TLS 1.3

for encrypted MQTT transport, plus an onboard Secure Element (SE) with cryptographic accelerator and secure key storage. Configure TLS in the MQTT broker and sensor firmware settings as needed for production deployments.

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Grafana Dashboard Access

Starting the Docker Stack

```
cd ~/zemfyre-sensor
docker compose up -d
```

Method 1 — Zemfyre Admin Panel *(Recommended)*

The Admin Panel at port **51850** is the intended end-user interface. It opens Grafana inside a fullscreen, clean iframe with no Grafana navigation bars or login prompts — exactly like a kiosk display.

- 1 Open a browser and navigate to `http://192.168.2.30:51850`
- 2 The Zemfyre Admin Panel loads with a dark navigation bar at the top showing the Zemfyre logo.
- 3 Click the **Dashboard** button in the top navigation bar.
- 4 The Grafana dashboard opens in a fullscreen iframe — no login required, 1-second auto-refresh active.
- 5 To switch **Sensor** chip at the top-left of **SENSOR_1** , **SENSOR_2** , etc. (see sensors, click ▼ the dashboard and Figures 1 & 2 the select below).

Why use the Admin Panel?

Port 51850 provides a cleaner experience: no Grafana header/sidebar, no sign-in screen, direct fullscreen dashboard view. It also gives access to

Settings

(time sync, database management) in one place.

Method 2 — Direct Grafana Access *(Advanced)*

Parameter	Value
URL	<code>http://192.168.2.30:53000</code>
Port	53000 (external) → 3000 (Grafana internal)
Authentication	Anonymous access enabled — no login required
Admin credentials	<code>admin</code> / <code>admin</code> (if login prompt appears)
Auto-refresh	1 second

Use direct Grafana access to create or edit dashboards, manage data sources, or explore the InfluxDB query builder.

Selecting a Sensor

The dashboard supports multiple sensors simultaneously. Click the **Sensor** ▾ chip at the top-left of the dashboard to open the selector — all panels instantly update to show data from the chosen device. Sensors appear as **SENSOR_1**, **SENSOR_2**, **SENSOR_3**, **SENSOR_4**, corresponding to IPs `192.168.2.100`, `.101`, `.102`, `.103`.



Figure 1 — Sensor selector dropdown (Admin Panel · port 51850). Click the **Sensor** ▾ chip (top-left) to open the list. Select **SENSOR_1** ... **SENSOR_4** to switch between connected devices. All gauges and history charts update immediately.



Figure 2 — Full dashboard view with SENSOR_1 active. Panels show live gauge + 5-minute history for Temperature (30 °C), Humidity (22%), Pressure (99.84 kPa), Proximity (4.60 mm), IAQ (Indoor Air Quality) and Ambient Light. Top-right: **DASHBOARD** and **SETTINGS** buttons are the Admin Panel navigation.

Metrics Displayed

Panel	Unit	Sensor / Source
Temperature	°C	BME688
Pressure	kPa / hPa	BME688
Humidity	%rH	BME688
IAQ (Indoor Air Quality)	score	BME688 (computed)
Ambient Light	lx	BH1750
Proximity	mm	VL53L4ED (Time-of-Flight)

Dashboard Troubleshooting

Problem	Solution
Admin panel not loading (port 51850)	Run <code>docker compose up -d</code> ; check <code>docker ps grep admin</code>
Connection refused (port 53000)	Run <code>docker ps grep grafana</code> ; restart with <code>docker compose up -d</code>
No data in panels	Confirm solid green LED; check MQTT: <code>mosquitto_sub -h 127.0.0.1 -p 1883 -t '#' -v</code>
Cannot reach Pi IP	Run <code>ping 192.168.2.30</code> — ensure PC is on the same subnet
No sensors in dropdown	Ensure sensors are powered and MQTT-connected (solid green LED)
Login screen shown	Use <code>admin</code> / <code>admin</code>

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Full CLI Command Reference

Getting Help

Type `help` for the main menu. Append `-h` to any command for detailed usage: `eth -h`, `bm688 -h`, etc.

6.1 System Commands

core — System Control

```
core -status      # System status overview
core -reset       # Apply config & reboot
core -version     # Firmware version
```

os — OS Diagnostics

```
os -get rt        # OS runtime in ms
os -get stats     # Scheduled task info
```

app — Application

```
app -state dia    # Enter diagnostics mode
app -state exit   # Exit / reset
app -display on   # Show readings in CLI
app -display off  # Hide output (default)
app -poll <ms>    # Set polling interval
app -poll get     # Get polling interval
```

tasks — Task List

```
tasks             # List active tasks
```

6.2 Network & MQTT Commands

```
# — Ethernet / network —————
eth -ifconfig                # Show current bindings
eth -ifconfig ip <x.x.x.x>   # Set device IP
eth -ifconfig gw <x.x.x.x>   # Set gateway
eth -ifconfig sn <x.x.x.x>   # Set subnet mask
eth -ifconfig dns0 <x.x.x.x> # Set primary DNS
eth -ifconfig dns1 <x.x.x.x> # Set secondary DNS
                             # (core -reset required after changes)

# — MQTT broker —————
eth -mqtt                    # Show MQTT binding
eth -mqtt ip <x.x.x.x>       # Set broker IP
eth -mqtt url "<broker-url>" # Set broker URL/hostname
eth -mqtt port <port>       # Set broker port (default 1883)
eth -mqtt stats              # Show publish statistics

# — Connectivity test —————
eth -ping <x.x.x.x>          # ICMP ping
```

6.3 Sensor Commands

sensor — Sensor Manager

```
sensor -list      # List all available sensors
sensor -status    # Initialization status per sensor
```

bm688 / bme688 — BME688 Environmental Sensor

```
bm688 -setoff temperature <val> # Temperature offset (°C)
bm688 -setoff pressure <val>     # Pressure offset (Pa)
bm688 -setoff humidity <val>     # Humidity offset (%rH)
bm688 -get offset                 # Show all offsets
```

adxl345 — Accelerometer

```
adxl345 -report set on  # Enable reporting (reset required)
adxl345 -report set off # Disable reporting
adxl345 -report get     # Get current state
```

vl53 — VL53L4ED Time-of-Flight Proximity

```
vl53 -setoff <mm>      # Set calibration offset in mm
vl53 -get offset        # Show all configured offsets
```

bh1750 — Ambient Light Sensor

```
bh1750 -h          # Show command help
```

6.4 EEPROM, Device & LED Commands

eeeprom — Non-Volatile Storage

```
eeeprom -erase      # Erase all EEPROM
eeeprom -default     # Erase & restore default IP bindings
eeeprom -pgerase <n> # Erase page n only (0 ≤ n ≤ 7)
```

Warning — Destructive

`eeeprom -erase` and `eeeprom -default` clear all network and MQTT configuration. Re-enter all settings and `core -reset` after running.

name — Device UID

```
name -set <name>    # Set custom device name
name -get            # Return device name
```

stsled — Status LED Control

```
stsled -r on/off/dc # Red channel on / off / duty-cycle
stsled -g on/off/dc # Green channel
stsled -b on/off/dc  # Blue channel
stsled -RGB 0xRRGGBB # 24-bit colour blend e.g. 0x00FF00
```

i2c — I2C Bus Diagnostics (Advanced)

```
i2c bus0 -if up/down # Interface up or down
i2c bus0 -read        # Current-index read
i2c bus0 -rread a=<addr> cl=<len> rl=<len> cmd=<d0:d1>
i2c bus0 -write       # Write operation
i2c bus1 <options>    # Same for bus 1
```

7 Sensor Validation & LED Status

ZUS801LP — 8-in-1 Measurement Reference

Measurement	Unit	Sensor	Notes
Temperature	°C	BME688	Ambient air temperature
Barometric Pressure	hPa / Pa	BME688	300–1100 hPa
Relative Humidity	%rH	BME688	0–100 %rH
Air Quality (VOC/CO ₂ eq)	Ω / AQI score	BME688	Gas resistance + computed AQI
Ambient Light	lx	BH1750	Illuminance level
Ambient Sound Level	dB / level	Onboard microphone	Acoustic environment monitoring
Proximity	mm	VL53L4ED	Time-of-Flight, object detection
3-Axis Acceleration	g	ADXL345	X, Y, Z axes

Verification Procedure

- 1 Check sensor initialization:

```
sensor -status
```
- 2 Enable live CLI output:

```
app -display on
# Output: GAS[x.xx], TEMPERATURE[x.xx], PRESSURE[x.xx], HUMIDITY
[x.xx]
```
- 3 View active system tasks:

```
tasks
```
- 4 Verify MQTT messages reach the broker:

```
mosquitto_sub -h 127.0.0.1 -p 1883 -t '#' -v
```
- 5 Open Grafana at <http://<Pi-IP>:51850> and confirm live data updates.

LED Status Summary

LED State	System State	Action
 Yellow toggling	No MQTT broker configured	Configure with <code>eth -mqtt</code>
 Green toggling	Configured; connecting	Wait; verify broker is running
 Green solid	Fully operational	No action needed
 Red (any)	Hardware / critical error	Check power; consider reflashing firmware

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Troubleshooting

LED Stuck Yellow

- Check MQTT broker IP/port: `eth -mqtt`
- Confirm Mosquitto running: `docker ps | grep mosquitto`
- Confirm port 1883 open: `ss -ltnp | grep 1883`
- Test reachability: `eth -ping <broker-ip>`
- After fixing, apply with `core -reset`

No Serial Connection in PuTTY

- Verify COM port in Device Manager → Ports
- Install correct driver (CH340, CP210x, or FTDI)
- Try baud rates: 9600, 57600, 115200
- Check physical cable and adapter connections

Network Settings Not Applied

- Always run `core -reset` after network changes
- Verify IP does not conflict with another device
- Check SPE cable connection and termination

Sensors Not Initializing

- MQTT must be configured before sensors start — configure and reset first
- Verify power supply meets specifications
- Run `sensor -status` to identify which sensors failed

Grafana Shows No Data

- Confirm solid green LED (MQTT connected)

- Listen for MQTT: `mosquitto_sub -h 127.0.0.1 -p 1883 -t '#' -v`
- Check Node-RED flow: `http://<Pi-IP>:51880`
- Check InfluxDB: `http://<Pi-IP>:58086`

Diagnostic Command Quick Reference

```
core -status      # System overview
core -version     # Firmware version
tasks            # Active task list
os -get stats     # OS task scheduling
eth -ifconfig     # Network settings
eth -mqtt        # MQTT binding
eth -mqtt stats   # Publish success/failure counts
eth -ping <ip>    # ICMP ping test
sensor -list      # Detected sensors
sensor -status    # Per-sensor init status
app -display on   # Stream live readings to CLI
```

Time Synchronisation Testing

```
sudo timedatectl set-ntp false
sudo systemctl stop systemd-timesyncd
sudo date -s "+5 minutes"      # Apply offset
date
# Re-enable after testing:
sudo systemctl start systemd-timesyncd
sudo timedatectl set-ntp true
```

9 Setup Checklist

Hardware

- ☐ Sensor power supply connected and power LED on
- ☐ SPE Ethernet cable connected
- ☐ USB-to-Serial adapter installed with correct driver
- ☐ PuTTY configured: Serial · correct COM port · 115200 baud
- ☐ Serial CLI accessible — `help` returns output

Network Configuration

- ☐ Raspberry Pi `eth0` on sensor subnet (e.g. 192.168.2.30/24)

- ☐ Sensor IP set: `eth -ifconfig ip 192.168.2.100`
- ☐ Gateway set: `eth -ifconfig gw 192.168.2.1`
- ☐ `core -reset` performed after network changes
- ☐ Ping successful: `eth -ping <sensor-ip>`

MQTT Configuration

- ☐ Docker stack running: `docker compose up -d`
- ☐ Mosquitto listening on port 1883
- ☐ MQTT broker IP configured: `eth -mqtt ip 192.168.2.30`
- ☐ MQTT port configured: `eth -mqtt port 1883`
- ☐ `core -reset` performed after MQTT configuration
- ☐ Status LED is solid  **green**
- ☐ MQTT messages visible with `mosquitto_sub`

Grafana Dashboard

- ☐ Grafana accessible at `http://<Pi-IP>:51850`
- ☐ Dashboard loads and shows sensor panels
- ☐ Temperature, Humidity, Pressure, Gas panels show live values
- ☐ Dashboard auto-refreshes every 1 second

Final Validation

- ☐ All 8 sensor measurements reporting data (temperature, pressure, humidity, air quality, light, sound, proximity, acceleration)
- ☐ MQTT publish stats clean: `eth -mqtt stats`
- ☐ All sensors initialized: `sensor -status`

Sensor
ZUS801LP · 8-in-1 · IEEE 802.3cg

SPE Technology
zemfyre.com

Technical support: zemfyre.com